

A VALORANT DATA PROJECT

Meta Data

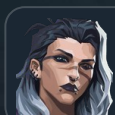
What would the pros pick?



Harun Tontus · Friday 25 September 2026

VALORANT for anyone who hasn't played

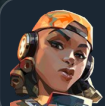
MAP: LOTUS



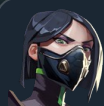
Fade



Omen



Raze



Viper



Vyse

A real pro line-up on Lotus

A PRO MATCH, BEST OF THREE

Map 1 chosen by team A

Map 2 chosen by team B

Map 3 the one left over, if needed

Map

The arena. Seven are in play at a time, and the set changes over the years.

Agent

A character with its own abilities. No two teammates pick the same one.

Line-up

A team's five agents. Roles make them follow patterns.

Players try to optimise for the best line-up

ME

Four years of competitive play, achieving one of the highest ranks. Viewer of the pro tour.

THE GAP

Stats sites show past pick and win rates. They don't test which agents work better together.

WHO IT'S FOR

Players picking a line-up, and fans curious what the pros would do.

Three questions

01

What would the pros pick?

A map and four agents in, the fifth out.

02

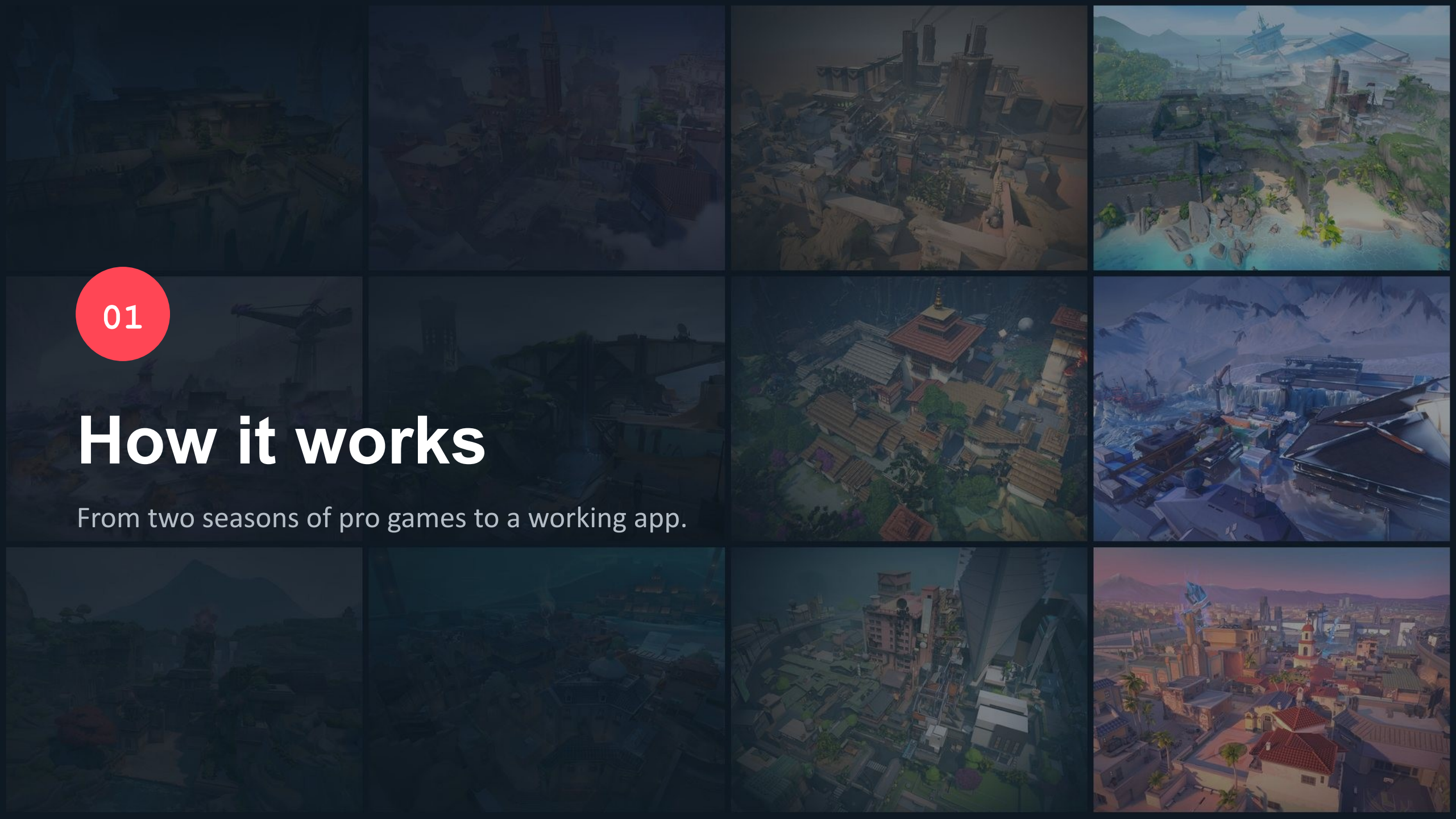
Do agent picks predict who wins?

Using only what's known before the game.

03

Do some line-ups over-perform?

Is a combination better than its parts?



01

How it works

From two seasons of pro games to a working app.

From pro games to an app, in six steps

1

Collect

2,376 pro games over two seasons

2

Clean

Fixed 3 errors that fail silently

3

Frame

Hide one agent: which completes the line-up?

4

Learn

Gradient boosting: which agents go together

5

Test

Week by week, earlier games only

6

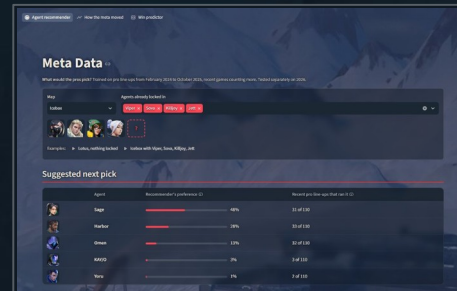
Show

A three-page Streamlit app

BUILT WITH

Python, pandas, scikit-learn,
Streamlit

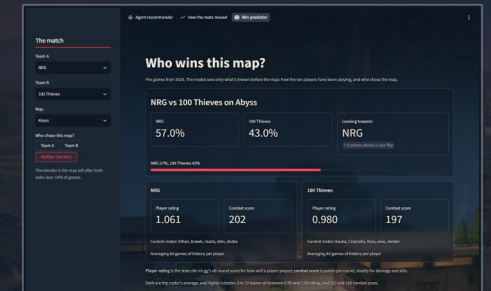
Data: Kaggle, from vlr.gg



Agent recommender



How the meta moved



Win predictor

The task

MAP: ICEBOX



Four agents locked in. What's the fifth?

ITS TOP 3 SUGGESTIONS



HOW IT'S SCORED

- Hide one agent from a real pro line-up. Is it in the top 3?
- The bar to beat: the most-played rule, whatever pros played most here lately.

The result

91.3%

rest of 2025

87.4%

2026: new season, new agents

PROS' PICK IN THE TOP 3, FOUR AGENTS LOCKED IN

Random guess



≈13%

Most-played rule, 2025



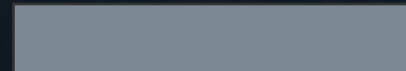
77.3%

Recommender, 2025



91.3%

Most-played rule, 2026



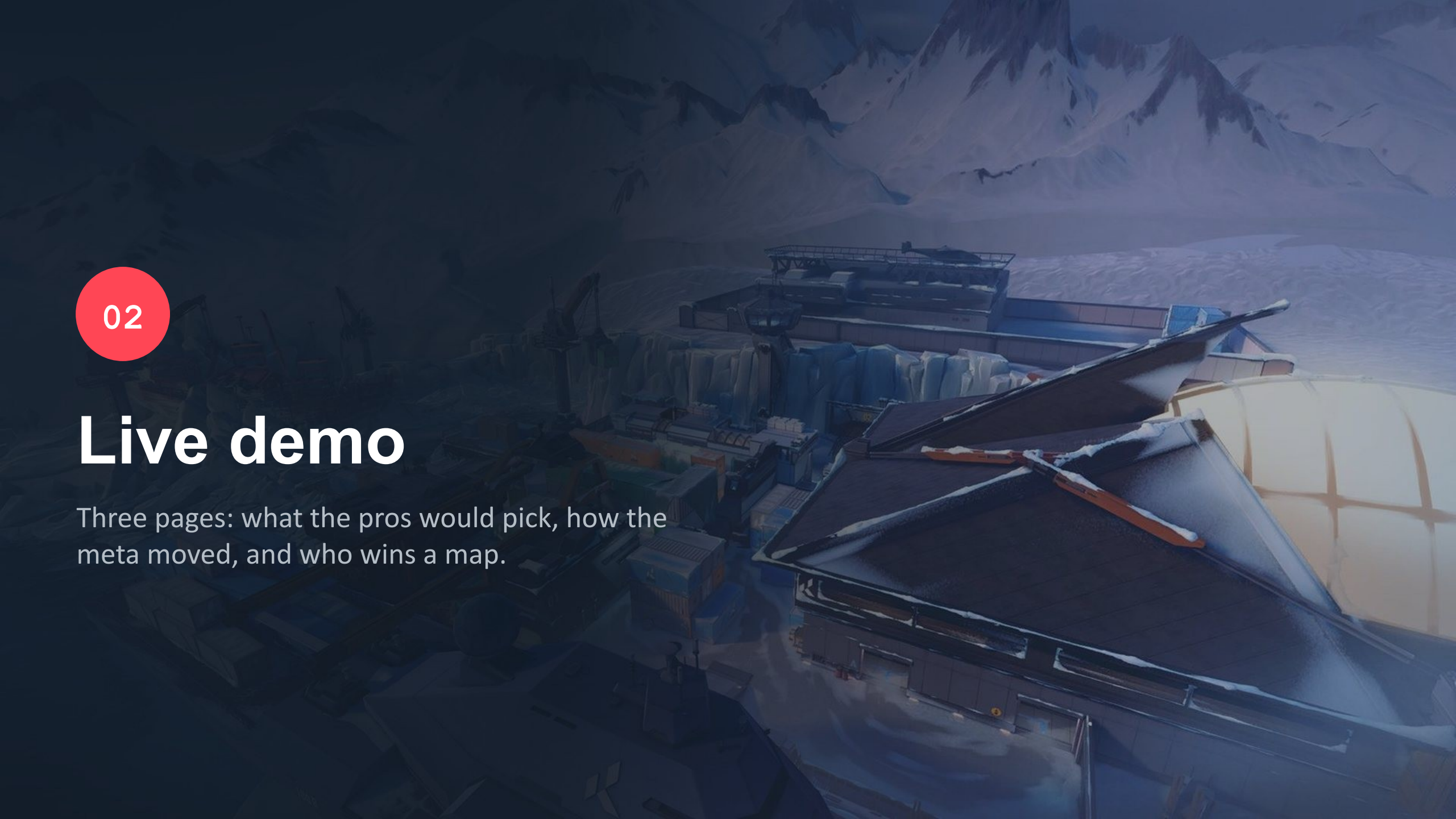
72.1%

Recommender, 2026



87.4%

+14 and +15 points over the most-played rule, even allowing for luck

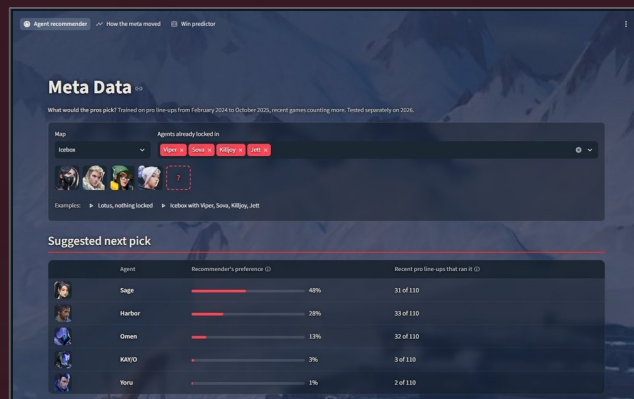


02

Live demo

Three pages: what the pros would pick, how the meta moved, and who wins a map.

What to watch for



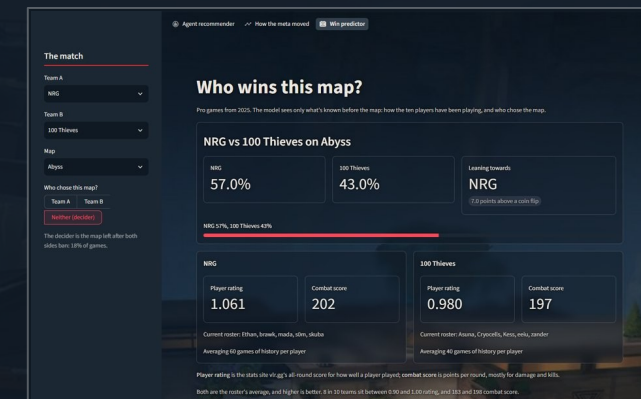
Agent recommender

Select a map and agents for recommendations



How the meta moved

Pick rates month by month, per agent



Win predictor

Testing whether agent picks change who wins

03

What I learnt

What surprised me, and what I'd improve next.



Four lessons I learnt

Split by time, not at random

A random split lets the model peek at the future meta.

The simple rule is hard to beat

'What pros played lately' already gets 77%.

Winners are far harder to predict

Best win model: 54.5%. 'Map picker wins': 52.9%.

Years of data can shrink to months

The game keeps changing: a year-old game counts under 1%.

What I'd improve

LIMITS TODAY

- Shows what pros pick, not what wins
- Weaker with fewer agents locked in
- Weaker on brand-new maps and agents

NEXT

- Keep it updated with new pro games
- Test it on ranked games
- An LLM in the app that suggests tactics for the chosen map and agents

Conclusion

With four agents locked in, the pros' pick is in its top three nearly nine times in ten, even on a new season.





Thanks for listening

Any questions?

Harun Tontus

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